

Task:

- Assemble a functional, organized, and safe mini rack; install and terminate a patch panel; create and test custom-length Ethernet cables; and connect, label, manage, and verify the rack's physical network infrastructure.

Student must demonstrate:

- Assemble the physical rack or configure a stable shelf setup according to the approved layout and equipment-clearance requirements.
- Mount or install the managed switch in the proper position, orienting it for service access, ventilation, and secure attachment.
- Mount or install the patch panel securely, align it with the rack or shelf layout, and leave adequate access for punching down or terminating cables.
- Position the mini PC so that it is stable, ventilated, accessible for maintenance, and protected from cable strain or accidental displacement.
- Create custom-length Ethernet cables using approved cable, modular plugs or patch-panel hardware, a cable stripper, crimping or termination tools, and the assigned wiring standard.
- Test each custom cable before installing it in the rack, recording the cable identifier and test result.
- Connect devices to the patch panel and connect the patch panel and other network devices to the correct switch ports according to the physical and logical diagrams.
- Route cables professionally using appropriate paths, bend radius, bundling, separation, and strain relief without obstructing ventilation or service access.
- Label both ends of every cable with clear, durable, and consistent identifiers that match the cable schedule, port assignments, and diagrams.
- Verify every connection by checking link lights, port assignments, cable seating, equipment stability, and physical security before declaring the rack complete.
- Mount or position equipment appropriately.
- Connect power and network cables correctly.
- Label both ends of each cable.

Observable criteria for proficiency:

- Every connection matches the approved physical diagram.
- Cables are labeled and routed neatly.
- No unnecessary cable tension or blocked airflow is present.
- Devices power on without unsafe behavior.
- The student can explain the signal path from the mini PC to the switch and the Off Production Network

Mini Rack Skill Check Rubric

Criteria	Exemplary (4)	Proficient (3)	Developing (2)	Beginning (1)
Diagram Adherence	Every connection precisely matches the approved physical diagram with no deviations.	All connections match the diagram with only minor, non-functional discrepancies.	Most connections match the diagram, but 1-2 significant errors are present.	Multiple connections do not match the diagram, or the diagram was not followed.
Cable Labeling & Routing	All cables are labeled at both ends with clear, consistent identifiers. Routing is professional, with optimal paths and proper bundling.	Cables are labeled at both ends and routed neatly, with minor inconsistencies in identifier format or routing.	Most cables are labeled, but some labels are missing or routing is somewhat disorganized.	Many cables lack labels, or routing is disorganized and unprofessional.
Airflow & Cable Management	No cable tension is present. Airflow is completely unobstructed. Proper bend radius and strain relief are maintained throughout.	Minimal cable tension is present. Airflow is mostly clear, with only minor obstructions that do not affect function.	Some cable tension or airflow blockage is present but does not immediately threaten equipment.	Excessive cable tension or significant airflow blockage could damage equipment or reduce performance.
Device Power & Safety	All devices power on immediately with no unsafe behavior. Connections are secure and equipment is stable.	Devices power on with only minor delays or require one adjustment. No safety concerns are present.	Devices eventually power on but require troubleshooting or show minor unsafe conditions.	Devices fail to power on properly or demonstrate unsafe behavior, such as overheating, sparking, or instability.
Signal Path Explanation	Student clearly and accurately explains the complete signal path from the mini PC through the switch to the Off Production Network using appropriate technical terminology.	Student accurately explains the signal path with minor omissions or imprecise terminology.	Student provides a partial explanation of the signal path but omits key components or shows confusion.	Student cannot explain the signal path, or the explanation contains major errors.

Total Points: ____ / 20

Performance Levels

- **Exemplary:** 17-20 points
- **Proficient:** 13-16 points
- **Developing:** 9-12 points
- **Beginning:** 5-8 points

Instructor Feedback: